

International Islamic University Chittagong (IIUC)

Department of Computer and Communication Engineering

Midterm Examination

Program: B.Sc (Engg.)
Course Code: CCE-3509
Total Marks: 30

Semester: Spring 2026
Course Title: Computer Architecture
Time: 1 Hour 30 Minutes

- (I) Answer all the questions. The figures in the right-hand margin indicate full marks.
(II) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1	Understand the Overview, Computer System, Arithmetic and logic, Central processing unit and parallel organization.
CLO2	Understand the Computer and Processor Design, Hazards; Exceptions; external and internal memory Pipeline and multiple processor systems.
CLO3	Develop and design an instruction set architecture and subsystems of central processing unit.

Bloom's Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Q1.	a)	Distinguish between Computer Architecture and Computer Organization.	CLO1	R	4
	b)	Briefly discuss the evolution of computers and how these changes impacted computer performance and usability.	CLO1	An	6
Q2.	a)	Define Multiprocessor. Explain the characteristics of multiprocessors connected by a network.	CLO1	U	4
	b)	Explain the role of cache memory in computer system. Illustrate cache organization of Pentium-4.	CLO3	An	6
Q3.	a)	What is virtual memory? Highlight different types of virtual memory.	CLO1	R	4
	b)	Demonstrate the working procedure of Direct Memory Access (DMA) on a computer system. What are the advantages of it?	CLO2	An	6

Or

Q3.	a)	Write short notes (Any two): i. Cache replacement policy, ii. DRAM, iii. Thunderbolt, iv. Volatile Memory	CLO1	R	4
	b)	Define Interrupt driven I/O. Analyse how it uses interrupt signal to improve CPU efficiency.	CLO2	An	6

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International Islamic University Chittagong (IIUC)

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Midterm Examination

Program: B.Sc (Engg.)
 Course Code: CCE-3509
 Total Marks: 30

Semester: Autumn 2025
 Course Title: Computer Architecture
 Time: 1 Hour 30 Minutes

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1	Understand the Overview, Computer System, Arithmetic and logic, Central processing unit and parallel organization.
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CLO3	Develop and design an instruction set architecture and subsystems of central processing unit.

Bloom's Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Q1.	a)	Explain the basic architecture of a computer system including its main components.	CLO1	An	5
	b)	Distinguish between Harvard Architecture and Von Neumann Architecture.	CLO1	R	5
Q2.	a)	What is a multiprocessor system? Explain the differences between symmetric and asymmetric multiprocessor.	CLO1	U	4
	b)	How does the cache memory work in a computer system? Briefly describe Pentium-4 cache organization.	CLO2	An	6
Q3.	a)	What is memory hierarchy in computer architecture? Explain the significance of it.	CLO2	R	5
	b)	Explain the working principles of Direct Memory Access (DMA) on a computer system. What are the advantages of it?	CLO2	An	5

Or

Q3.	a)	Write Short Notes (any two): (i) Solid State Drive (ii) Thunderbolt (iii) Interrupt (iv) Register	CLO1	R	2
	b)	Define Programmed I/O. Discuss different types commands in Programmed I/O.	CLO2	U	5
	c)	What is BUS? Shortly explain the types of buses used in computer architecture?	CLO1	R	3

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Midterm Examination

Program: B.sc (Engg.)
 Course Code: CCE-3509
 Total Marks: 30

Semester: Spring-2025
 Course Title: Computer Architecture
 Time: 1 Hours 30 Minutes

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

- CLO1** Understand the Overview, Computer System, Arithmetic and logic, Central processing unit and parallel organization.
CLO2 Understand the Computer and Processor Design, Hazards; Exceptions; external and internal memory Pipeline and multiple processor systems.
CLO3 Develop and design an instruction set architecture and subsystems of central processing unit.

Bloom's Levels of the Questions

Letter Symbols
 Meaning

R U Ap An E C
 Remember Understand Apply Analyze Evaluate Create

Q1.	a)	Explain the basic structure of a computer system including its main components.	CLO1	U	4
	b)	Describe the difference between arithmetic and logic operations with examples.	CLO1	U	3
	c)	Define the term "Central Processing Unit (CPU)". List its major parts.	CLO1	R	3
Q2.	a)	Explain the role of buses in a computer system, including the functions of the data bus, address bus, and control bus. How do these components contribute to the interaction between the processor and other hardware?	CLO1	U	5
	b)	Explain with an example why modern systems often prefer serial communication over parallel despite its slower theoretical data rate.	CLO2	Ap	5
Q3.	a)	What is a multiprocessor system? Briefly explain the difference between symmetric and distributed multiprocessor architectures.	CLO1	U	4
	b)	Consider a system with four processors sharing memory and I/O resources. Design a simple block diagram showing symmetric multiprocessing (SMP) and explain how tasks are handled and resources are shared. Then, describe how the same would differ in a distributed multiprocessing system.	CLO2	Ap	6
OR					
Q3.	a)	What is the memory hierarchy in computer architecture? Explain its importance in terms of access speed and cost.	CLO1	U	4
	b)	Describe how cache memory works in a computer system. Explain what happens when the CPU requests data — include the concepts of cache hit, cache miss, and how data is fetched from memory. Use a simple example to support your explanation.	CLO2	Ap	6

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Department of Computer and Communication Engineering

Midterm Examination

Program: **B.Sc. (Engg.)**
 Course Code: **CCE -3509**
 Total Marks: **30**

Semester: **Spring 2024**
 Course Title: **Computer Architecture**
 Time: **1 Hour 30 Minutes**
 Date: **29/02/2024**

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

- CLO1** Understand the Overview, Computer System, Arithmetic and logic, Central processing unit and parallel organization.
- CLO2** Understand the Computer and Processor Design, Hazards; Exceptions; external and internal memory Pipeline and multiple processor systems.

Letter Symbols Meaning		Bloom's Levels of the Questions								
		R Remember	U Understand	Ap Apply	An Analyze	E Evaluate	C Create			
Q1.	a)	Explain the contrast between computer architecture and computer organization.						CLO1	An	4
	b)	What is a SIMD computer? Explain its architecture using a block diagram.						CLO1	U	6
Q2.	a)	What is the significance of the address bus in the 8085 μ P, and how does it differ from the data bus?						CLO1	U	4
	b)	Draw a neat diagram and explain the function of the program counter (PC) register in a common bus system for basic computer registers, and how it interacts with other components.						CLO1	An	6
Q3.	a)	Using one sentence each, explain the meaning of cache hit, cache miss and mapping in terms of cache memory.						CLO2	U	4
	b)	Consider a machine with a byte addressable main memory of 2^{32} bytes divided into blocks of size 32 bytes. Assume that a direct mapped cache having 512 cache lines is used with this machine. i. What is the size of the TAG field in bits? ii. What is the size of the OFFSET field in bits? iii. What is the size of the INDEX (LINE) field in bits? Using a schematic diagram, show which parts of the address serve as offset, index and tag fields.						CLO2	An	6
OR										
Q3.	a)	Explain the hierarchical memory system in terms of their access times and sizes.						CLO2	U	4
	b)	Consider a 4-way set associative cache consisting of 128 lines with a line size of 64 words. The CPU generates a 20-bit address of a word in main memory. i. What is the size of the TAG field in bits? ii. What is the size of the OFFSET field in bits? iii. What is the size of the INDEX (LINE) field in bits? Using a schematic diagram, show which parts of the address serve as offset, index and tag fields.						CLO2	An	6

*** Good Luck ***

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Midterm Examination

Program: B.Sc. (Engg.)
Course Code: CCE -3509
Total Marks: 30

Semester: Spring 2023
Course Title: Computer Architecture
Time: 1 Hour 30 Minute
Date : 18/02/2023

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.
ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

1. Understand the Overview, Computer System, Arithmetic and logic, Central processing unit and parallel organization.
2. Understand the Computer and Processor Design, Hazards; Exceptions; external and internal memory Pipeline and multiple processor systems.

Letter Symbols Meaning	Bloom's Levels of the Questions						C
	R	U	Ap	An	E		
	Remember	Understand	Apply	Analyse	Evaluate		Create
a) Differentiate between computer architecture and computer organization.						CLO1	An
b) What is meant by the 'stored program' concept? Draw a diagram of Von-Neumann's architecture and explain it briefly.						CLO1	R, U
a) What are the limitations of Programmed I/O?						CLO1	R
b) Explain the working of DMA using a suitable block diagram. Give an example application of DMA data transfer.						CLO1	E
a) Explain the role of cache memory in a computer.						CLO2	I
b) Using a suitable block diagram, explain the architecture of a SIMD computer. Identify at least two applications for which SIMD computers are suitable.						CLO2	I

OR

- a) Why are memories organized in the hierarchy? Which are the factors to be considered while selecting a particular memory type? CLO2
- b) Explain the importance of each type of memory with a diagram of the memory hierarchy. CLO2

***** Good Luck *****